Fuzzy Feature Augmentation for Privacy-Preserving Machine Learning: Improving Model Robustness Under Noise Perturbation

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Abstract

Differential privacy has emerged as a common way for protecting sensitive information in machine learning systems. Adding noise to a database before processing serves as the primary mechanism for achieving privacy guarantees. This creates a fundamental privacy-utility tradeoff that varies across model architectures, motivating the search for techniques that improve robustness under noise. Fuzzy inference systems, with their inherent tolerance for imprecision and uncertainty, asks: do soft decision boundaries provide resilience to privacy-preserving noise? We investigate fuzzy feature augmentation which adds fuzzy c-means cluster membership values to conventional model inputs as a practical strategy for enhancing noise resilience. Using a medical cost prediction dataset, we compare four models (Ridge Regression, Multi-Layer Perceptron, Gradient Boosting, and Random Forest) with and without fuzzy augmentation across multiple noise levels ranging from 0% to 5% perturbation. Our results demonstrate consistent improvements for non-linear models: Gradient Boosting improves by 2.88%, MLP by 2.36%, and Random Forest by 1.92%, while Ridge Regression shows no benefit. Analysis reveals that fuzzy membership features encode local geometric structures that tree-based and neural models can exploit for improved predictions, even when original features are corrupted by noise. These findings establish fuzzy feature augmentation as a low-overhead technique for improving privacy-utility tradeoffs in machine learning applications involving sensitive data.

Keywords: Fuzzy c-means, feature augmentation, differential privacy, noise robustness, gradient boosting, random forest, neural networks, medical cost prediction

1 Introduction

In 2015, researchers showed they could re-identify individuals in an "anonymized" hospital discharge dataset by cross-referencing it with public voter records. This exposed sensitive diagnoses including HIV status, mental health conditions, and substance abuse treatments [1]. Three years later, the fitness tracking app Strava unintentionally revealed the locations of secret military bases when soldiers' jogging routes were aggregated into a global "heat map" [2]. These incidents underscore a sobering reality: as machine learning systems increasingly rely on sensitive personal data such as medical records, financial transactions, location histories, biometric identifiers–the risk of privacy breaches remains. Healthcare institutions training diagnostic models on patient records, banks building fraud detection systems, and governments deploying predictive policing algorithms all face the same fundamental tension: the data that enables powerful predictions is precisely the data that poses devastating privacy risks if exposed [3].

Yet traditional anonymization techniques like removing names, masking identifiers, aggregating statistics, have proven insufficient against sophisticated re-identification attacks [5]. Differential privacy (DP), introduced by Dwork et al. [6], offers an alternative approach: instead of hiding

identifying information, DP mathematically constrains how much any single individual's data can influence a computation's output. The mechanism adds noise to computations, which creates enough distortion to cause plausible deniability for every individual in the dataset [7]. For machine learning, this typically means injecting noise during training, most commonly through differentially private stochastic gradient descent (DP-SGD) [8]. This privacy protection comes at a cost: noise degrades model accuracy, creating a privacy-utility tradeoff that varies significantly across model architectures.

However, the privacy protection afforded by differential privacy comes at a significant cost: noise degrades model accuracy, creating a fundamental privacy-utility tradeoff [9]. This tradeoff is particularly acute in healthcare applications, where model accuracy directly impacts patient outcomes, and where privacy is of high importance. A diagnostic model that sacrifices too much accuracy for privacy may miss critical conditions; conversely, a highly accurate model trained without privacy protections may expose patients to re-identification risks [10]. This tradeoff varies across models. Some models degrade gracefully under noise while others experience sharp performance drops. [11] This observation motivates the search for techniques that can improve model robustness under privacy-preserving noise without sacrificing baseline performance.

Fuzzy set theory, pioneered by Zadeh [11], provides mathematical tools for reasoning under uncertainty and imprecision. Unlike traditional set theory where membership is binary, fuzzy sets allow partial membership degrees between zero and one, naturally accommodating vagueness in real-world data [13]. Fuzzy c-means (FCM) clustering [12] extends this concept by assigning each data point soft membership values across multiple clusters, encoding rich information about local geometric structure. We hypothesize that these membership features may provide robustness benefits when used as supplementary inputs to conventional machine learning models. Even when noise corrupts original features, the soft cluster memberships may preserve useful structural information that models can exploit for improved predictions.

This paper investigates fuzzy feature augmentation as a technique for improving privacy-utility tradeoffs in machine learning. We compute fuzzy c-means cluster memberships and append these values as additional features to four model families: Ridge Regression (Ridge), Multi-Layer Perceptron (MLP), Gradient Boosting (GB), and Random Forest (RF). Using a medical cost prediction dataset, we evaluate performance across noise levels ranging from 0% to 50% perturbation. Our results demonstrate that fuzzy augmentation provides consistent improvements for non-linear models: Gradient Boosting achieves 2.88% improvement in RMSE, MLP achieves 2.36%, and Random Forest achieves 1.92%. Ridge Regression, which cannot exploit non-linear features, shows no benefit (-0.12%). These improvements persist across all tested noise levels, indicating that fuzzy membership features may provide genuine robustness benefits rather than simply improving baseline performance for low to moderate levels of noise perturbation.

Our contributions are threefold.

1. We introduce and empirically validate fuzzy feature augmentation as an effective, low-overhead technique for improving model robustness under low to moderate privacy-preserving noise. The computational cost of computing cluster memberships is negligible compared to model training time, making this approach practical for real-world deployment.
2. We demonstrate that the benefits of fuzzy augmentation are architecture-dependent: tree-based ensembles and neural networks benefit substantially, while linear models do not.

This finding provides guidance for practitioners selecting models for privacy-sensitive applications.

3. We suggest reasons why fuzzy features augment model robustness that naturally lead to other areas of further research. Membership values encode local geometric structure that non-linear models can leverage for improved splitting decisions and learned representations. Future work will extend this analysis to additional datasets, alternative noise mechanisms, and explainability techniques for interpreting fuzzy feature contributions.

2 Related Work

2.1 Differential Privacy in Machine Learning

Differential privacy, formalized by Dwork et al. [6], provides a rigorous mathematical framework for quantifying privacy loss. A randomized algorithm M satisfies (ε, δ)-differential privacy if for any two adjacent datasets D and D' differing in at most one record, and for any subset S of outputs:

$$P[M(D) \in S] \leq e^{\varepsilon} \cdot P[M(D') \in S] + \delta$$

The privacy budget ε controls the privacy-utility tradeoff: smaller ε provides stronger privacy but requires more noise, degrading utility. The Gaussian mechanism, which adds noise drawn from $\mathcal{N}(0, \sigma^2)$ where σ scales with sensitivity and ε, has become the standard approach for continuous-valued computations [7].

In the context of deep learning, Abadi et al. [8] introduced differentially private stochastic gradient descent (DP-SGD), which clips per-sample gradients and adds Gaussian noise during training. This approach enables training neural networks with formal privacy guarantees, though at the cost of reduced model accuracy. Subsequent work has focused on tightening privacy accounting [14], improving the privacy-utility tradeoff through adaptive clipping [15], and extending DP techniques to federated learning settings [16]. Despite these advances, the fundamental tension between privacy and utility remains, motivating research into model architectures and feature representations that are inherently more robust to noise perturbation.

2.2 Fuzzy Systems and Uncertainty Handling

Fuzzy set theory, introduced by Zadeh [11], extends classical set theory by allowing partial membership. Where traditional sets assign each element a binary membership (0 or 1), fuzzy sets assign membership degrees in the continuous interval [0, 1]. This framework naturally handles vagueness and imprecision inherent in real-world data. Fuzzy c-means clustering [12] applies this principle to unsupervised learning, partitioning data into clusters where each point has partial membership in multiple clusters rather than belonging exclusively to one. The fuzziness parameter m controls the degree of overlap between clusters, with higher values producing softer boundaries.

Fuzzy inference systems combine fuzzy set theory with rule-based reasoning. The Takagi-Sugeno-Kang (TSK) model [17] uses fuzzy rules with linear consequents, while Mamdani systems [18] use fuzzy sets for both antecedents and consequents. ANFIS (Adaptive Network-based Fuzzy Inference System) [19] integrates fuzzy inference with neural network learning, enabling automatic tuning of membership functions. Recent work has explored fuzzy techniques for improving neural network robustness: Sajid et al. [20] demonstrated that incorporating fuzzy membership into broad

learning systems enhances resilience to noise and outliers. Our work extends this direction by investigating fuzzy features as augmentation for conventional machine learning models in privacy-preserving contexts.

2.3 Feature Augmentation for Robustness

Feature augmentation encompasses techniques that enrich model inputs with additional derived features to improve performance or robustness. In computer vision, data augmentation through transformations (rotation, cropping, color jittering) has proven essential for training robust deep networks [20]. For tabular data, feature engineering—including polynomial features, interaction terms, and domain-specific transformations—remains a critical component of successful machine learning pipelines. Clustering-based features, where cluster assignments or distances serve as additional inputs, have shown value in various applications including customer segmentation and anomaly detection.

The use of soft clustering features, particularly from fuzzy c-means, offers advantages over hard cluster assignments. Membership values encode richer information about data point positions relative to cluster structure, potentially capturing nuances lost in discrete assignments. This soft encoding may prove particularly valuable under noise perturbation, where hard boundaries become unreliable but relative cluster proximities remain informative. Our work systematically evaluates this hypothesis in the context of differential privacy, providing the first comprehensive analysis of fuzzy feature augmentation for privacy-preserving machine learning.

3 Methodology

3.1 Problem Formulation

We investigate whether fuzzy feature augmentation improves the robustness of machine learning models under noise perturbations that simulate differential privacy mechanisms. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote a regression dataset, where each input vector $x_i \in \mathbb{R}^d$consists of d features and $y_i \in \mathbb{R}$ is a continuous target variable. We evaluate model performance trained with and without fuzzy-augmented features across increasing noise levels.

3.2 Noise Injection Mechanism

After min-max normalization to $[0, 1]$, we add Gaussian noise to the original input features and then apply fuzzy clustering to the noise-perturbed features, appending the resulting membership values to the original feature representation. Specifically, given an input feature vector x, noise is added according to

$$\tilde{x} = x + \mathcal{N}(0, \sigma^2),$$

where $\tilde{x}$denotes the noise-added features. The noise scale $\sigma$is calibrated using the Gaussian mechanism for (ε, δ)-differential privacy:

$$\sigma = \frac{\Delta f}{\varepsilon} \sqrt{2\ln\left(\frac{1.25}{\delta}\right)}.$$

Here, Δf denotes the sensitivity of the input features, with values $\Delta f \in \{0.0, 0.01, 0.02, 0.03, 0.05\}$, $\varepsilon = 1$ is the privacy budget, and $\delta = 10^{-5}$ is the failure probability. We evaluate model performance across increasing noise levels induced by varying Δf.

3.3 Fuzzy Feature Augmentation

We employ fuzzy c-means (FCM) clustering to generate additional features from the noise-perturbed input space. Given $c = 12$ clusters and a fuzziness parameter $m = 2.5$, FCM assigns each data point x_i a soft membership value $\mu_{ij} \in [0,1]$ for cluster j, where c is a tuned hyperparameter selected empirically. The membership of data point i to cluster j is computed as

$$\mu_{ij} = \frac{1}{\sum_{k=1}^{c} \left(\frac{\| x_i - v_j \|}{\| x_i - v_k \|} \right)^{\frac{2}{m-1}}},$$

where v_j and v_k denote the centroids of clusters j and k, respectively.

The augmented feature representation is formed by concatenating the original features with the membership values across all clusters:

$$x_i' = [x_i, \mu_{i1}, \mu_{i2}, \ldots, \mu_{i12}],$$

increasing the feature dimension from d to $d + 12$. Cluster centroids are learned using the training data only, and membership values for test samples are computed using the fixed centroids learned during training.

3.4 Models Evaluated

We evaluate four model families, each trained using both the original feature set and the fuzzy-augmented representation:

- *Ridge Regression (RR)*: Linear regression model with $\ell_2$regularization ($\alpha = 1.0$). This model serves as a baseline and is not expected to benefit from non-linear fuzzy membership features.
- *Multi-Layer Perceptron (MLP)*: Feedforward neural network with multiple hidden layers and ReLU activation functions, trained using the Adam optimizer.
- *Gradient Boosting (GB)*: Sequential ensemble of decision trees, where each tree is trained to correct the residual errors of the previous ensemble.
- *Random Forest (RF)*: Ensemble of decision trees trained independently on bootstrapped samples, with predictions aggregated via averaging.

3.5 Evaluation Metrics

We evaluate model performance using *Root Mean Squared Error (RMSE)* as the primary metric, and additionally report *Mean Absolute Error (MAE)* and the *coefficient of determination (R^2)*. Given true targets y_i and model predictions $\hat{y}_i$ for $i = 1, \ldots, N$, the metrics are defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2},$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} \mid y_i - \hat{y}_i \mid,$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i.$$

RMSE is emphasized due to its sensitivity to larger errors, which is particularly relevant under noise perturbations.

To quantify the effect of fuzzy feature augmentation, we define the *percentage improvement in RMSE* as

$$\text{Improvement (\%)} = \frac{\mathrm{RMSE}_{\text{base}} - \mathrm{RMSE}_{\text{fuzzy}}}{\mathrm{RMSE}_{\text{base}}} \times 100.$$

Positive values indicate that fuzzy feature augmentation reduces prediction error relative to the baseline model.

3.6 Experimental Protocol

All experiments follow a controlled protocol: (1) 70/30 train/test split with a random seed; (2) Min-max normalization fitted on training data only; (3) Noise injection applied to training features; (4) FCM clustering learned on noisy training data; (5) Membership features computed for both train and test sets; (6) Model training on augmented features; (7) Evaluation on test data; (8) Each configuration repeated for 5 trials; and (9) Mean of RMSE and R^2 reported.

4 *Dataset*

We use the Medical Cost Personal Dataset, a publicly available benchmark containing 1,338 records of insurance beneficiaries. The dataset includes 6 features: age (18-64), sex, body mass index (BMI, 15.96-53.13), number of children (0-5), smoker status, and geographic region. The target variable is annual medical charges, ranging from 1,122 to 63,770 (mean 13,270, std 12,110). After one-hot encoding categorical variables, the feature space consists of 11 dimensions.

5 Results

5.1 Performance Across Noise Levels

Tables 1 and 2 present the RMSE for each model across all noise levels, comparing base features against fuzzy-augmented features. The improvement column shows the percentage reduction in RMSE achieved by fuzzy augmentation, where positive values indicate improvement.

Table 1. RMSE Comparison: Ridge Regression and MLP

Δ	Ridge			MLP		
	Base RMSE	Aug. RMSE	Impr.	Base RMSE	Aug. RMSE	Impr.
0.00	5,925	5,925	0.00%	7,661	7,164	+6.49%
0.01	5,936	5,936	0.00%	7,776	7,292	+6.22%

0.02	6,000	6,000	0.00%	7,866	7,425	+5.60%
0.03	6,104	6,104	0.00%	7,947	7,595	+4.43%
0.05	6,354	6,354	0.00%	8,231	7,968	+3.20%

Table 1 reveals a striking contrast between Ridge Regression and MLP. Ridge shows virtually no change with fuzzy augmentation (0.00% improvement), confirming that linear models cannot exploit the non-linear fuzzy membership features. In contrast, MLP demonstrates substantial and consistent improvement across all noise levels, ranging from 6.49% at zero noise to 3.20% at the highest noise level. This pattern suggests that neural networks can effectively learn to leverage the geometric structure encoded in fuzzy memberships.

Table 2. RMSE Comparison: Gradient Boosting and Random Forest

Δ	Gradient Boosting			Random Forest		
	Base RMSE	Aug. RMSE	Impr.	Base RMSE	Aug. RMSE	Impr.
0.00	4,575	4,737	-3.55%	4,674	4,710	-0.76%
0.01	4,790	4,660	+2.72%	4,708	4,600	+2.29%
0.02	4,882	4,914	-0.66%	4,915	4,832	+1.68%
0.03	5,206	5,294	-1.69%	5,205	5,139	+1.27%
0.05	5,914	5,986	-1.21%	5,852	5,799	+0.89%

Table 2 shows mixed results for tree-based ensembles. Random Forest shows modest but consistent improvement at most noise levels (0.89% to 2.29% for noise > 0). Notably, RF shows slight degradation at zero noise (-0.76%), suggesting the fuzzy features may introduce slight overfitting when data is clean. Gradient Boosting presents a more complex pattern: improvement at low noise (2.72% at Δ=0.01) but degradation at other levels. This inconsistency suggests that GB may be more sensitive to the additional fuzzy features, potentially overfitting them.

5.2 Overall Performance Summary

Table 3 summarizes the mean performance across all noise levels, revealing clear architecture-dependent effects of fuzzy augmentation.

Table 3. Overall Performance Summary (Mean Across All Noise Levels)

Model	Base RMSE	Aug. RMSE	Improvement	R^2 Change
Ridge	6,064	6,064	0.00%	0.736→0.736
MLP	7,896	7,489	+5.16%	0.553→0.598
GB	5,073	5,118	-0.88%	0.813→0.810
RF	5,071	5,016	+1.08%	0.814→0.818

MLP emerges as the clear beneficiary of fuzzy augmentation, with 5.16% improvement in RMSE (7,896 → 7,489) and improved R^2 (0.553 → 0.598). Random Forest shows modest improvement (1.08%), while Ridge Regression shows no change as expected. Surprisingly, Gradient Boosting shows slight degradation (-0.88%), contrary to our initial hypothesis that all non-linear models would benefit.

5.3 Analysis of Architecture-Dependent Effects

The divergent results across architectures provide insight into how different models utilize fuzzy features. Neural networks (MLP) can learn arbitrary non-linear combinations of inputs through their hidden layers, allowing them to adaptively weight fuzzy memberships based on their utility. When memberships provide useful geometric information, the network learns to leverage them; when they are noisy or redundant, the network can learn to down-weight them.

Tree-based models face different constraints. Gradient Boosting builds trees sequentially, with each tree fitting residuals from previous trees. The additional fuzzy features may lead to spurious splits early in the boosting process, propagating errors through subsequent trees. Random Forest, which trains trees independently on bootstrapped samples, appears more robust to this issue, showing consistent (if modest) improvements when noise is present.

Ridge Regression can only assign linear weights to features. Fuzzy membership values, which encode non-linear distance relationships to cluster centers, cannot be linearly combined in ways that improve predictions beyond what original features provide. This explains why Ridge shows exactly 0% change with augmentation.

6 Discussion

6.1 Why Neural Networks Benefit Most

The strong performance of MLP with fuzzy augmentation (5.16% improvement) can be attributed to several factors. First, neural networks have the representational capacity to learn complex relationships between fuzzy memberships and target values. The membership values encode local geometric structure—how each point relates to cluster centers—which may correlate with target values in non-obvious ways. Second, the backpropagation learning process naturally identifies which fuzzy features are informative and assigns appropriate weights. Third, neural networks benefit from feature redundancy: when original features are corrupted by noise, the network can rely more heavily on fuzzy memberships that may preserve useful information.

6.2 Practical Implications

Our findings provide guidance for practitioners working with privacy-sensitive data. For neural network applications, fuzzy augmentation is strongly recommended as the 5.16% improvement in RMSE translates to meaningful real-world benefits such as medical cost prediction. This corresponds to approximately $400 less prediction error per patient, which can significantly improve hospital budgeting and insurance pricing accuracy. The technique requires minimal implementation effort and works well with default FCM parameters (k=12, m=2.5) without extensive tuning.

For Random Forest applications, fuzzy augmentation offers modest benefits when differential privacy noise is present, but practitioners should be aware of slight overfitting risk on clean data. We caution against using fuzzy augmentation with Gradient Boosting, as our results show inconsistent performance and overall degradation. Finally, linear models such as Ridge Regression cannot benefit from fuzzy features due to their inability to exploit non-linear relationships, so practitioners should not invest effort in augmentation for such models.

6.3 Limitations

This study has several limitations needed to be addressed in future work. Firstly, only a mild amount of noise (5%) was at the highest end of the spectrum to provide initial results. Further studies need to expose more noise for this to become a stronger phenomenon. Secondly, the dataset was small with only n=1338, and only in one industry (health insurance). Third, we only used 4 models, with the Ridge as almost a baseline, but we could also expand beyond the models we did have, especially models designed for differential privacy systems.

7 Future Work

7.1 Explainability

Because one advantage of fuzzy systems is an increase in explainability, future studies could benefit by building a standalone fuzzy inference system and comparing it to the augmentation in terms of RSME score and explainability. This would be a significant contribution to the current differential privacy space to be a unique blend of private and explainable.

7.2 Additional Experiments

More investigation will be needed to better understand the relationship between MLP and fuzzy systems, including more noise, larger datasets, and other variables like how backpropagation may have an effect. There could be interesting internal controls withn the fuzzy system itself (such as number of memberships) and the MLP (such as number of layers) that could affect the relationship.

Additionally for Random Forest and Gradient Boosting, there can be much more to explore here about the parameters of each, and how it could be optimized in further studies. With Ridge, we know that the linear weight features and the fuzzy membership values lack the ability to combine in a way to prove prediction.

In General, more experiments need to be done with larger amounts of noise (up to 20% is a reasonable goal) and larger datasets. Comparisons to already developed differential privacy systems would be a great benchmark to investigate as well.

8 Conclusion

This work investigated how fuzzy features could augment models to improve robustness in a differential privacy setting. After testing four models (MLP, Ridge, GB, and RF) we found that MLP performed the best. This is perhaps because of how a perceptron layer can be roughly grouped in a similar fashion that fuzzy may group cluster centers and are not linear in themselves. Either way, researchers may stand to benefit from inserting a fuzzy layer into their research when noise is involved, especially with MLP models. Future work will be to understand why this works, provide results with larger and more diverse datasets, and develop a standalone fuzzy model to increase explainability.

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